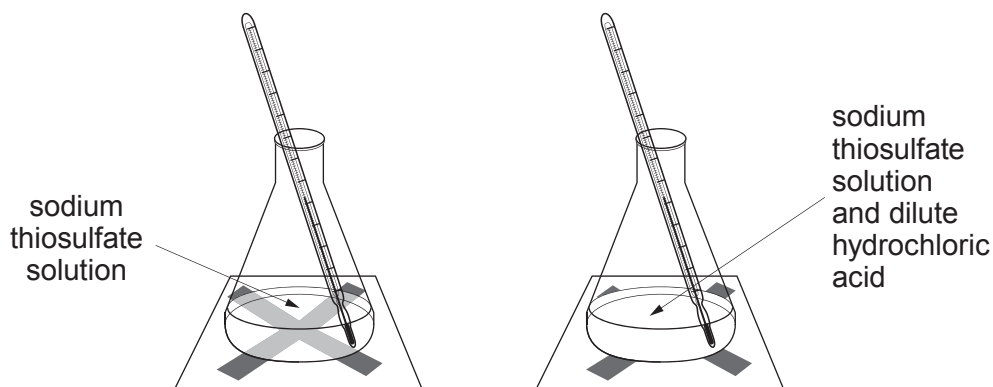


Examiner
only

5. The chemical industry is developing new processes to manufacture chemicals more efficiently.
- (a) (i) When sodium thiosulfate reacts with hydrochloric acid the solution becomes cloudy.



Describe a method used to investigate how temperature affects the rate of the reaction between hydrochloric acid and sodium thiosulfate solution. Include details of apparatus and volumes of chemicals used in your answer. [6 QER]

(ii) State the **two** variables that need to be controlled in this experiment. [2]

1

2

(iii) State **one** reason why these variables need to be controlled. [1]

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(b) In a different experiment hydrochloric acid was poured into a flask containing different volumes of sodium thiosulfate solution and water. The time taken to react was measured. The results are shown in the table below.

Volume of acid (cm ³)	Volume of sodium thiosulfate (cm ³)	Volume of water (cm ³)	Time for cross to disappear from view (s)
60	55	5	44
60	50	10	51
60	45	15	57
60	40	20	83
60	35	25	106

(i) State what conclusion can be made from the results. [1]

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(ii) Explain the results in terms of particles. [2]

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